Database Systems CoSc3051_

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Database Systems

- Data: known facts that can be recorded with implicit meaning
- Database: collection of related data
- Database Management System (DBMS): tool for creating and managing databases.
- A file system lacks features of a database:
  - Data redundancy and inconsistency,
  - Difficulty in accessing data,
  - Data isolation,
  - Integrity problems,
  - Atomicity problems,
  - Concurrent access anomalies.

Database Systems

- Database Approach Characteristics:
  - Self-describing nature: complete definition or description of the database structure and constraints
  - Data abstraction and Insulation between programs and data
  - Support of multiple views of the data: view is not stored
    - subset of the database or
    - contain virtual data derived from the database
  - Sharing of data and multiuser transaction processing

Database Systems

- Data model: collection of concepts that can be used to describe the structure of a database
  - High-level or conceptual: ER model, OO model
  - Low-level or physical: details on how data is stored, for specialists
  - Representational (or implementation): for end user understanding but can be implemented directly

Database Systems

- Database schema: description of a database, doesnt change frequently, specified during database design.
  - Three schema Architecture
    - Internal level has an internal schema: physical storage structure, complete details
    - Conceptual level has a conceptual schema: describe whole database for users; entities, data types, relationships, user operations and constraints…
    - External or view level: part of the database that a particular user group is interested in

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- Data Independence: change the schema at one level
  - Logical data independence: change the conceptual schema without having to change external schemas
  - Physical data independence: change the internal schema without having to change the conceptual schema

Database Systems

- Database modeling:
  - Phases - database development life cycle (DDLC)
    - Requirement Analysis: most important step, what is needed?
    - Conceptual Database Design
    - Logical database design (Data model mapping) - tables, indexes, views, transaction, access privileges
    - Physical Design - data files, table spaces

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- Database modeling - ERD (Entity Relationship Diagram)
  - Entity and Attribute
    - Entity may be an object with a physical existence, has attributes (properites that describe it)
    - Attribute types:
      - Composite vs Simple (Atomic): composite can be divided to subparts while simple cant.
      - Single-valued vs Multivalued: single number vs array of numbers
      - Derived: age determined from birth date
      - Key: one or more distinct attributes
        - Super key: collectively identifies an entity
        - Candidate key: minimal super key
        - Primary key: one of candidate keys for uniquely identifying entity set

Database Systems

- Database modeling - ERD (Entity Relationship Diagram)
  - Relationship: association between entity types
    - Unary: one entity (i.e. employee manages employee)
    - Binary: two entities (i.e. student takes course)
    - Cardinality:
      - One to one (1-1): each entity take part only once in the relationship
      - One to many (1-M): one entity set can take part only once in the relationship set and entities in other entity set can take part more than once in the relationship set
      - Many to many (M-M): all entity sets can take part more than once in the relationship

Database Systems

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- Database modeling - Enhanced Entity Relationship (EER) Model
  - improvements or enhancements to the existing ER to reduce complexity
  - includes the concepts of:
    - subclass and superclass,
    - categories (UNION types),
    - attribute and relationship inheritance

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- Relational Model Concepts in DBMS
  - Attribute: Each column in a Table
  - Tables: has two properties rows and columns. Rows represent records and columns represent attributes.
  - Tuple: a single row of a table containing a single record
  - Column: set of values for a specific attribute
  - Relation Schema: name of the relation with its attributes

Database Systems

- Properties of Relations
  - Name of the relation is distinct from all other relations.
  - Each relation cell contains exactly one atomic (single) value.
  - Each attribute contains a distinct name.
  - Each tuple has no duplicate value.
  - Order of tuple can have a different sequence.

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- Functional Dependency: constraint between two attributes or sets of attributes of entities
  - denoted by $X \rightarrow Y$: values of Y component depend on, or are determined by the values of X component.
  - Types:
    - Partial Dependency: an attribute which is not a member of the key is dependent on some part of the key (composite/candidate key)
    - Full Dependency: functionally dependent on that attribute and not on any of its proper subset
    - Transitive Dependency: If A functionally governs B, AND If B functionally governs C THEN A functionally governs C

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- Database Anomalies: flaws in databases which occurs because of poor database design
  - Insertion Anomaly: entire primary key is not known, database cannot insert a new record properly
  - Deletion Anomaly: a record is deleted that results in the loss of other related data
  - Update Anomaly: a change to one attribute value causes the database to either contain inconsistent data or causes multiple records to need changing.

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- Normalization: organizing the data in database to avoid data redundancy and anomalies:
  - Normal forms: maintenance rules that are used to eliminate or reduce redundancy:
    - First normal form (1NF): all underlying domains of attributes contain only atomic (simple or indivisible) values

Database Systems

- Normalization: organizing the data in database to avoid data redundancy and anomalies:
  - Normal forms: maintenance rules that are used to eliminate or reduce redundancy:
    - Second normal form (2NF): it is in 1NF and remove any partially dependent attributes and place them in another relation (only full dependency with key is allowed)
      - i.e. Emp-Teams(EmpId, Name, BDate, Gender, TeamId, Project, TeamName)
      - decomposition
        - EmpId → {Name, BDate, Gender}
        - TeamId → {Project, TeamName}
      - 2nf:
        - Employees(EmpId, Name, BDate, Gender)
        - Teams(TeamId, Project, TeamName)
        - Emp-Teams(EmpId, TeamId)

Database Systems

- Normalization: organizing the data in database to avoid data redundancy and anomalies:
  - Normal forms: maintenance rules that are used to eliminate or reduce redundancy:
    - Third Normal Form (3NF): it is in 2NF and fully functionally dependent on every key of R and non-transitively dependent on every key of R..
      - i.e. zip code and addresses of employee to separate table with foreign key from employee to address

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- Attribute Data Types:
  - CHAR - fixed-length character
  - VARCHAR() - variable-length characters
  - DATE – actual date
  - BLOB – binary large object (good for graphics images, audio, sound clips, etc.)
  - NUMERIC - positive/negative numbers

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- Attribute Data Types:
  - Exact
    - SMALLINT → 16 bit
    - INT → 32 bit
    - LONG → 64 bit
    - DECIMAL → 128 bit
  - Approximate
    - FLOAT (single precision) → 32 bit with ≈ 6-9 digit precision...
    - REAL(variation of double) → 32 bit with ≈ 15 digit precision...
    - DOUBLE (double precision) → 64 bit with ≈ 32 bit precision...

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- Data/File Storage
  - Primary storage: store a data which can be operated directly by the CPU; RAM and cache
  - Secondary storage: cannot be processed directly by the CPU, must first be copied into primary storage; hard disk
  - Tertiary storage: removable media, offline storage for archiving databases; CD-ROMs, DVDs

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- File Organizations: how the file records are physically placed on the disk
  - Heap file (unordered file): no particular order, appending new records at the end of the file
  - Sorted file (sequential file): ordered by the value of a particular field (sorting key)
  - Hashed file: hash determines record's placement on disk, hash is applied on a field (hash key)
  - Secondary organization (auxiliary access structures):

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- Indexes
  - Dense Index: created for every search key value (for each records) in each block; index contains search key value and a pointer to the actual record… faster, more size
  - Sparse Index: created only for some of the data records.
  - Single Level Ordered Indexes: assumed that a file already exists with some primary organization; index access structure is usually defined on a single field of a file, called an indexing field

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- Indexes
  - Single Level Ordered Indexes:
    - Primary: specified on the ordering key field of an ordered file of records. every record has a unique value for that field.
      - Primary index is an ordered file whose records are of fixed length with two fields.
        - PK: ordering key field
        - pointer: points to disk block address

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- Indexes
  - Single Level Ordered Indexes:
    - Clustering: have the same value for the clustering field, if ordering field is not a key field - same values across records - it can be used as clustering key.
      - Clustering index is also an ordered file with two fields.
        - Clustering field of the data file
        - Disk block/cluster pointer
    - Secondary: some primary access already exists, secondary index is dense, faster retrieval.
      - Many secondary indexes can be created for the same file each represents an additional means of accessing that file based on some specific field.

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- Relational algebra: basic set of operations for the relational model, to specify basic retrieval requests (queries)
  - result of an operation is a new relation
  - basic operations
    - Selection (σ) - Selects a subset of rows from relation.
    - Projection (π) - Deletes unwanted columns from relation.
    - Cross-product (x) - Allows us to combine two relations.
    - Set-difference (–) - Tuples in relation 1, but not in relation 2.
    - Union (U) - Tuples in relation 1 and in relation 2.
    - Intersection (∩) - Only tuples in relation 1 and in relation 2 (only the common tuples).

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- SQL (Structured Query Language): statements for data definitions, and updates, both DDL and DML
  - DDL (data definition language)
    - CREATE, DROP, ALTER, TRUNCATE and RENAME
  - DML (data manipulation language)
    - SELECT, PROJECT, JOIN, INSERT, DELETE, and UPDATE